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ORIGINAL ARTICLE

Looking for non-hermaphrodite cacti: multidisciplinary studies in *Gymnocalycium bruchii* endemic to central Argentina

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Abstract

Key message **Through a multidisciplinary study we found that *Gymnocalycium bruchii*, an endemic cactus from central Argentina, acts as a dioecious species, which is the first record in this genus.**

Abstract Cactaceae species are typically hermaphroditic; however, about 2% have other different reproductive systems. These non-hermaphroditic species may develop sexual dimorphism in flowers or other reproductive, vegetative or ecological traits, besides a specific breeding system and floral ontogeny. Therefore, multidisciplinary research is necessary to fully understand reproduction in those species. For this purpose, we studied *Gymnocalicium bruchii*, a globose cactus endemic to central Argentina that is presumably dioecious or gynodioecious. We made observations in two natural and two cultivated populations. We made morphological observations of plants and flowers, and performed quantitative analyses to determine the sex ratio, size of plants and flowers, flower production, fruiting, among other variables. We performed hand-pollination, self-fertilization and free-pollination tests to determine the breeding system. Finally, we studied the anatomy and ontogeny of the reproductive organs using permanent histological slides of flower morphs at different stages. Our results confirm that *Gymnocalicium bruchii* is a dioecious species. Female flowers have atrophied anthers and a functional gynoecium that produces fruits and seeds. Male flowers are bigger and have a functional androecium but a sterile gynoecium. In the cultivated population, the sex ratio was 1/1, whereas the number of male individuals was higher in both natural populations. Pollination tests corroborated dioecy. Ontogenetic studies revealed that in female flowers the anthers collapse before microspore maturation, while in male flowers the gynoecium shows normal development of the ovary, style, stigma, and ovules; however, the latter are never fertilized.

Keywords Dioecy · Sexual dimorphism · Female:male ratio · Self-incompatibility · Floral development

Introduction

Cactaceae Juss. is a modern plant family that arose around 30 million years ago associated with global aridization processes (Arakaki et al. 2011). It has about 1850 species (Nyffeler and Eggli 2010), mainly distributed in arid areas of the Americas, characterized by long dry periods and ephemerals pulses of rainfall (Buhanan and Briggs 2010). As a consequence, species developed a general ecological strategy of stress tolerance, showing specific modifications such as slow growth rate, succulent stems and protective spines (Buhanan and Briggs 2010; Aliscioni et al. 2021; Perotti et al. 2022).

Regarding their reproductive system, species are typically hermaphrodite, with perfect flowers that present both a functional gynoecium and androecium, but with different degrees of self-compatibility (Pimienta-Barrios and del

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Castillo 2002; Martínez-Peralta et al. 2014; Fernández et al. 2020; Delgado-Ramírez et al. 2021). Species with non-hermaphrodite flowers are uncommon in the family. Only 24–26 species (ca. 2%) present dioecy or some intermediate stage, like gynodioecy, subdioecy or trioecy (Orozco-Arroyo et al. 2012; Callejas-Chavero et al. 2021). A few dioecious species are found in the genera *Pereskia* Mill., *Opuntia* Mill., *Echinocereus* Engelm., *Consolea* Lem., and gynodioecious species are known in *Mamillaria* Haw., *Opuntia* Mill. and other genera (ie. Fleming et al. 1994; Díaz and Cocucci 2003; Orozco-Arroyo et al. 2012; Sánchez and Vázquez-Santana 2018; Callejas-Chavero et al. 2021). This distribution suggests that dioecy has evolved independently in several cactus lineages.

Non-hermaphroditic breeding systems usually entail the appearance of different floral characteristics, leading to sex dimorphism in the primary and secondary sexual traits (Callejas-Chavero et al. 2021). The most important primary sexual trait is the differential development of the gynoecium or androecium, whereas among the secondary ones are variation in floral size, color, and nectar production (Fleming et al. 1994; Hernández-Cruz et al. 2018; Díaz and Coccuci 2003; Callejas-Chavero et al. 2021). The presence of dioecy, gynodioecy or trioecy could also have effects on other plant reproductive, vegetative or ecological traits, such as sex ratio, total production of flowers per sex, and size of individuals (Hoffman 1992; Fleming et al. 1994). In general, female plants need to allocate more resources to fruit and seed production, which could negatively affect plant size and flowering (Wilson and Harder 2003). Thus, female plants are expected to be smaller and to produce fewer and smaller flowers than their male counterparts (Wilson and Harder 2003).

The impossibility of self-fertilization in non-hermaphroditic flowers gives pollinators an important role, since sexual reproduction depends on the effective transport of viable pollen from the anthers of male flowers to the stigma of female flowers (or bisexual flowers in case of trioecious or gynodioecious sexual systems). Pollination treatments have been conducted in several genera of hermaphrodite cacti to corroborate the breeding system (Martínez-Peralta et al. 2014; Mandujano et al. 2014). However, a few studies have been conducted on the breeding system in gynodioecious species (Díaz and Cocucci 2003; Callejas-Chavero et al. 2021), dioecious and hermaphroditic *Echinocereu*s Engelm. species (Hernández-Cruz et al. 2018), dioecious *Cylindropuntia wolfi* (Ramadoss et al. 2022), and subdioecious *Consolea spinosissima* (Strittmatter et al. 2002), among others; in all cases, the different tests were fundamental tools to understand the particular reproductive processes of each species.

Ontogenetic studies of gynoecious or dioecious species are essential to know the precise timing of sexual differentiation and the underlying mechanisms. Although studies on the anatomy and development of flowers and fruits are common in Cactaceae (Fernandez et al. 2020), few studies have included non-hermaphrodite species (Strittmatter et al. 2002, 2006, 2008; Flores-Rentería et al. 2013; Sánchez and Vázquez-Santana 2018; Hernández-Cruz et al. 2018, 2019). Among the most recent and important works are those carried out on the gynodioecious species *M. dioica* K. brandegee (Sánchez and Vázquez-Santana 2018), dioecious populations of *E. coccineus* Engelm., *E. polyacanthus* Engelm., *E. pacificus* Engelm. (Haage) and *E. mombergerianus* G. Frank, hermaphrodite populations of *E. triglochidiatus* Engelm. (Hernández-Cruz et al. 2018), and the dioecious species *O. robusta* H. L. Wend. Ex Pfeiff. (Hernández-Cruz et al. 2019). In all these species, similar anatomical characteristics were observed in the abortive anthers of the female flowers, while the gynoecium of male flowers presented different degrees of atrophy, depending on the species. It should be noted that all those studies involved species from the Northern Hemisphere.

Gymnocalycium Pfeiff. ex Mittler is one of the most important South American cactus genera, with about 50 species, being the mountains of northwestern Argentina the richest area (Charles 2009; Meregalli et al. 2010; Demaio et al. 2011). The genus is likely monophyletic, with diploid or tetraploid species, with $2n = 22$ or $2n = 44$ (Martino et al. 2022; Bauk et al. 2022). It is also very popular among plant enthusiasts because of its beautiful flowers; for this reason, it is cultivated worldwide. Nevertheless, little is known about the reproductive biology of *Gymnocalycium*, which is typically considered a genus with hermaphroditic flowers (Charles 2009; Meregalli et al. 2010; Giorgis et al. 2015). However, based on floral morphology, some species show a supposed sexual dimorphism. Individuals with apparently perfect flowers (normally developed gynoecium and androecium) and others with apparently female flowers (with well-developed gynoecium and abortive anthers, devoid of pollen) have been observed (DE Gurvich, personal observations). The presence of these two types of flowers suggests that such species could be either gynodioecious or dioecious. These morphological characteristics were observed in *G. bruchii* (Speg.) Hosseus, *G. uruguayense* (Arechav.) Britton and Rose (M. Wick, personal communication), *G. gibbosum* (Haw.) Pfeiff. ex Mittler (Wolfgang Papsch, personal communication) and *G. monvillei* (Lem.) Pfeiff. Ex Briton and Rose (D.E. Gurvich, personal observations). However, to date, no *Gymnocalycium* species has been studied regarding its breeding system and floral ontogeny.

Therefore, our work aimed to study the reproductive biology of *G. bruchii*, an endemic species from the Sierras de Córdoba in central Argentina whose flowers present dimorphism. For this purpose, we made morphological observations of plants and flowers in two natural and two cultivated populations. We carried out breeding system experiments in cultivated plants and studied the morphoanatomy and ontogeny of the observed flower morphs. With this interdisciplinary study, we aim to gain a comprehensive understanding of the sexual reproduction of the species as well as to lay the groundwork for future studies that attempt to elucidate the evolution of the breeding systems of the whole genus.

Materials and methods

Studied species

Gymnocalycium bruchii (Speg.) Hosseus is a species endemic to the Sierras de Córdoba and San Luis in central Argentina (Charles 2009; Gurvich et al. 2014). The plant is globose, small, and presents a taproot; it usually forms colonies through vegetative growth (Fig. 1A). The species flowering early in spring and individuals produce a few flowers per season, which open quite synchronically (Giorgis et al. 2015).

Cultivated seed-grown plants

To analyze the reproductive system, we cultivated plants from seeds that were obtained from a population located in the Sierras Chicas (Gurvich et al. 2022). Plants were sown in 2014, in an open garden in Salsipuedes city, Córdoba province; blooming started in 2017. From the plants obtained, 30 healthy thirty individuals with reproductive capacity were selected and were carefully observed throughout their reproductive period, between August and November from 2017 to 2021. For each individual, we recorded the type of flowers developed, according to their exomorphological characters (female and apparently hermaphrodite). The sex of all individuals was checked during the study period to investigate whether sexual traits are fixed or can change over time, and if the different floral morphs produce fruits. Five flowers per morph were collected and cut longitudinally to observe and corroborate their sexual traits.

In 2020, the following variables we recorded for each individual: plant body diameter, date and time of

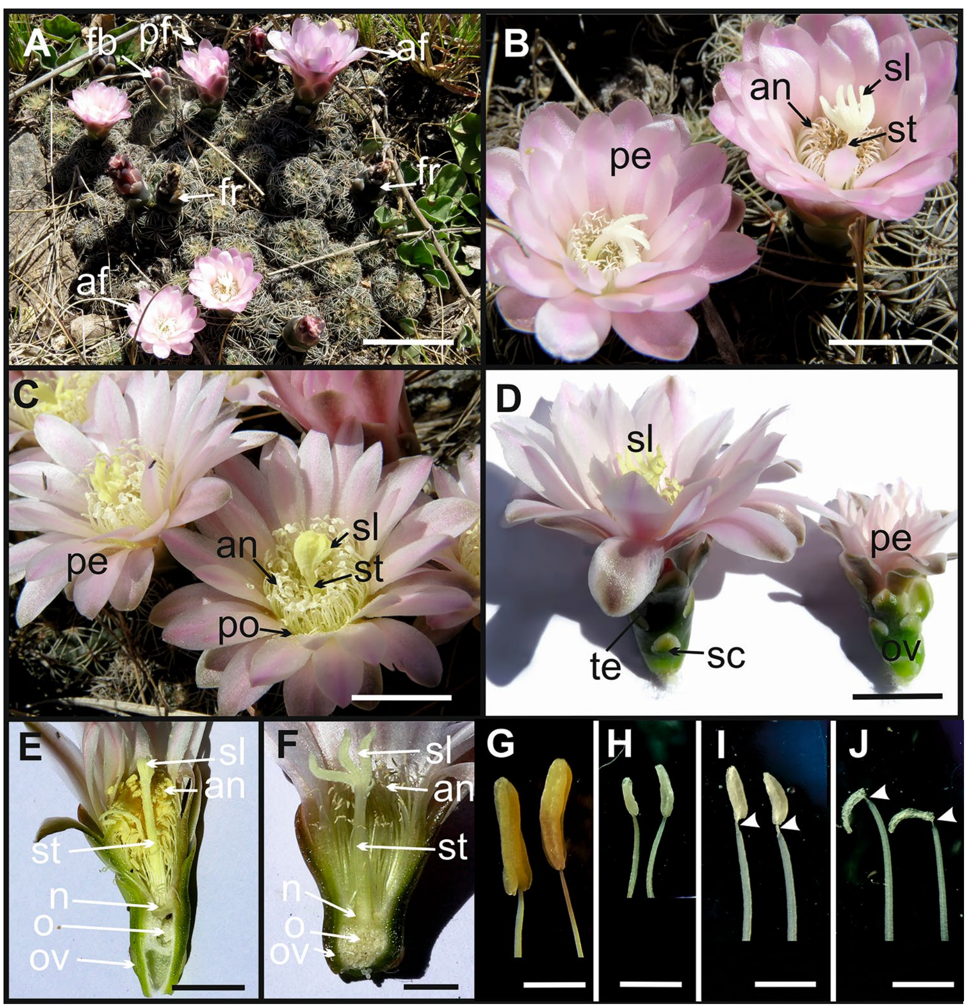

Fig. 1 Plants and flowers of *Gymnocalycium bruchii.* **A** Female plants with flowers at different developmental stages. **B** Female flowers at anthesis, with collapsed anthers. **C** Male flowers at anthesis, with dehiscent anthers. **D** Size comparison between male (left) and female (right) flowers. **E** Longitudinal section of male and **F**: Female flowers, showing the reproductive organs. **G** Stamens of male flowers, anthers normally developed, before dehiscence. **H** Stamens of male flowers with dehiscent anthers. **I** Stamens of female flowers, anthers underdeveloped, filament with constriction. **J** Stamens of female flowers at anthesis, with complete collapsed anthers, filament with constriction. Abbreviations: an: anther, af: anthesis flower, fb: flower bud, fr: fruit, n: nectary, o: ovule, ov: ovary, pe: perianth, pf; pre-anthesis flower, po: pollen, sc: scale, sl: stigma lobel, st: style, te: tepal. Scale: **A**: 30 mm. **B**–**E**: 10 mm. **F**: 5 mm. **G**–**J**: 1 mm

flowering, type of flowers produced, number of flowers produced per individual, and diameter of the perianth at anthesis.

We also carried out different pollination experiments to determine the sexual behavior of each floral morph and its breeding system. We analyzed the same 30 plants, 15 with female flowers and 15 with apparently hermaphrodite flowers. The experiments were carried out in an open garden. First, five plants with buds of each type were covered with a tulle bag to prevent the visit of pollinators and to verify if flowers can automatically self-fertilize. Then, to record the effects of cross-pollination, the flowers of five plants from each type were hand-pollinated with pollen from another apparently hermaphrodite plant. Finally, five plants were left as controls and exposed to natural/open pollination. After the flowering period, we recorded if fertilization was successful in each flower and if they produced fruits.

Natural populations

In 2019 we analyzed two populations in Córdoba, Argentina: one in the Sierras Chicas mountains, Camino del Cuadrado (31°07′ S, 64°25′ W), which was visited on September 25, and the other in the Sierras del Norte Mountains, San Pedro Norte (30°19′ S, 64°12′ W), which was visited on October 2. In each population, we analyzed 100 individuals and checked the presence of both types of flowers according to their exomorphological characters (female or apparently hermaphrodite morphs). We also recorded the number of branches, diameter of plant body, number of buds, flowers, and fruits, and perianth diameter at anthesis. Ten flowers per morph were collected and cut longitudinally to observe and corroborate their sexual traits.

Specimens of the Córdoba botanical garden collection

To establish if other populations of this species present the same floral traits, we analyzed plants in the Córdoba Botanical Garden, Córdoba, Argentina. This garden maintains a collection of *G. bruchii* plants from different localities. There are plants from the southern distribution limit of the species (Calmayo and Los Reartes populations, 32 01S 64 27W and 31 53S, 64 35W, respectively). A total of 13 individuals were analyzed throughout their flowering period, between August and September 2019. We recorded the date of flowering, floral morphs (female or apparent hermaphrodite), and fruit production by the individuals. Overall, we studied populations that cover a large area of the species' geographical distribution.

Morphoanatomy analyses

We performed laboratory studies to analyze the morphoanatomy and development of gynoecium and androecium of both floral morphs. With that aim, apparently female and hermaphrodite flowers at different stages were collected from the seed-grown plants in 2019. We use 10 flowers for each morph, of different sizes, from small 5 mm blossoms to open flowers, to try to reconstruct completely the ontogeny. The material was preserved in FAA (10 formaldehyde, 5 acetic acid, 50 ethanol, 35 water). Permanent slides of cross sections were made using classical histological techniques (Zarlavsky 2014). The material was dehydrated in a series of ethyl alcohol/xylene and included in paraffin. Serial sections approximately 10 µm thick were performed using a rotation microtome; then they were stained with astral blue-basic fuchsin and mounted with Canada balsam (Kraus et al. 1998; Zarlavsky 2014). Images were taken using a light microscope Primo Star-Carl Zeiss and a digital camera Nikon Coolpix 5200.

Data analyses

The character values of plants and flowers were statistically analyzed to determine if there were significant differences between floral types and among populations. We ran one-way ANOVAs and Tukey tests to analyze significant differences ($p \leq 0.05$) in some floral attributes, such as number of flowers, perianth diameter, and individual's body diameter, between the floral types in each population, because the environmental conditions in each site can have an effect on sex parameters. We also assessed differences across plants of different floral types among populations. All analyses were performed with the R software version 4.0.2 (R Core Team 2022).

Results

Reproductive traits

All morphological analyses confirmed that *G. bruchii* is a dioecious species, with individuals producing female flowers (Fig. 1A, B) that set fruits (Fig. 1A) and other individuals with male flowers (Fig. 1C). The species is not gynodioecious, since the apparently hermaphrodite flowers are functionally male. Therefore, morphs are hereafter referred to as female and male flowers.

Flowering starts in August and fruits reach maturity in October (Fig. 1A). Anthesis is diurnal, floral opening begins at 11 am and the maximum opening is between 12 and 2 pm. Flowers begin to close at 3 pm and complete closing occurs at 5:30 pm. Flower lifespan reaches an average of 3–4 days.

Both types of flowers have a colorful perigonium (Fig. 1B, C), composed of several cycles of light pink tepals, and green scales at the base of the flower and on the pericarpel. The flowers of both morphs are easily differentiated by their size, since the male flowers are bigger than the female ones (Fig. 1D). All flowers present an inferior ovary with numerous ovules (Fig. 1E, F), a developed style and numerous (5–10) papillate stigmatic branches. The stamens are very numerous, fused to the perigonium wall at different levels (Fig. 1E, F). Male flowers have a functional androecium (Fig. 1E), the anthers are bitecate and basifixed (Fig. 1G), with longitudinal dehiscence (Fig. 1H). Female flowers present an atrophied, sterile androecium (Fig. 1F). Anthers are not fully developed and do not reach normal size; the filament has a constriction just below the anther attachment (Fig. 1I) and, the anther collapses completely before opening (Fig. 1J).

All populations analyzed (two in the field, one at the Córdoba Botanical Garden, and our cultivated plants) show the same pattern: each individual produces either male or female flowers, with a marked sexual dimorphism. Since we analyzed populations from the extreme geographic locations of their distribution, we confirm that the whole species is dioecious. The seed-grown plants maintained the same type of flowers for at least the 6 years of study, and the male flowers never produce fruit, suggesting that in *G. bruchii* sex expression would be genetically determined, stable, and fixed over time.

Quantitative traits

The morphological character values of plants and flowers analyzed in each population, and the differences found between sexes are summarized in Table 1 and SM 1. In the field, the number of male individuals was higher than the number of female plants, with a female/male ratio of 0.64 in San Pedro Norte and of 0.48 in Camino del Cuadrado. However, in the cultivated plants, the number of male individuals was equal to the number of females, i.e. the ratio of female/male plants was 1. Each individual had a large number of branches in the natural populations, which was similar in both sexes. Cultivated plants only had the main axis of the body, probably because they were relatively young. Body diameter was similar between male and female plants in natural populations and only showed significant differences ($p = 0.046$) in the cultivated ones, with males being bigger (Fig. 1D). Flower diameter showed significant differences between female and male plants in Camino del Cuadrado ($p = 2.26e{-}11$) and cultivated plants ($p = 5.68e{-}07$). In the two populations, male flowers were considerably larger than female ones, evidencing a clear sexual dimorphism. No significant differences were found in the San Pedro Norte population, presumably due to the low number of flowers measured (Table 1). Mean number of flowers produced per individual was similar between female and male individuals in all populations, i.e. flower production was independent of sex. Finally, in the field, no male individuals with fruits were found, whereas in cultivated populations male flowers never fruited. Conversely, female plants of all the studied populations yielded fruit and seed.

The most important quantitative variables and the differences between populations are shown in Fig. 2 and SM 2. The population of San Pedro Norte had less developed individuals, since number of branches (Fig. 2A) and body diameter (Fig. 2B) were smaller than in the individuals from the Camino del Cuadrado. The cultivated individuals only presented one branch, but body diameter was bigger than in the plants of both natural populations (Fig. 2B). These results suggest that size of the individuals would depend on the environmental conditions where they occur. Flowers diameter was similar in all populations (Fig. 2C); therefore, this variable was found to depend on sex of flowers rather than on environmental conditions. On the other hand, the number of flowers produced per individual (Fig. 2D) was notably higher in cultivated plants than in the natural populations, whereas the population of San

Table 1 Mean character values of plant and flower of male and female individuals from three populations of *Gymnocalycium bruchii*

Sex	Camino del Cuadrado		San pedro norte		Cultivated plants	
	Male	Female	Male	Female	Male	Female
Individuals	71	34	61	39	15	15
Branches	7.3 (7.2)	6.2 (9.1)	4.1 (5.3)	2.4 (3.3)	1	1
Body diam mm	22.7 (5.8)	22.1 (4.2)	18.4 (3.3)	17.7 (4.2)	32.7 (5.2)*	27.6 (8.4)
Flower diam mm	30.1 (6)*	21.9 (4)	31 (6.3)	26.3 (1.8)	36.2 (6.6)*	26.2 (4.2)
Flower production	256	102	109	55	64	76
Flowers/individual	3.6 (3.8)	3 (3.1)	1.8 (1.4)	1.4 (0.8)	4,27 (2)	5.1 (2.4)
Fruit production	0	30	0	32	0	25

Comparative statistical analysis between female and male individuals of a single population. Each variable is expressed as the mean with its standard deviation in parentheses

*Indicates significant differences between sexes according to analysis of variance at $p < 0.05$

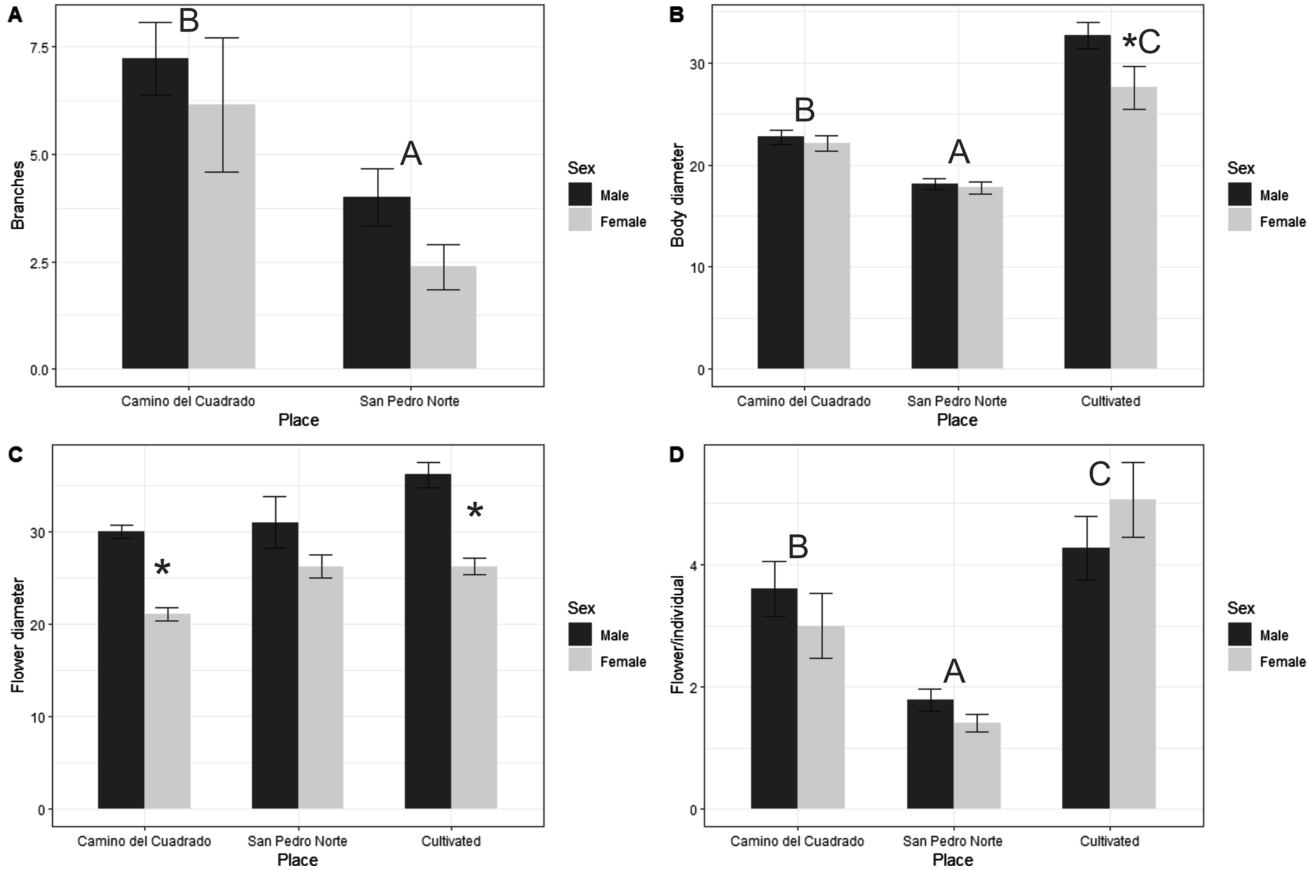

Fig. 2 Plant and flower traits for each sex and populations of *Gymnocalycium bruchii*. **A** mean number of branches or arms per individual. **B** mean body diameter expressed in mm. **C** mean flower diameter in fully open flowers expressed in mm. **D** mean production of buds, flowers, or fruits per individual. Comparative statistical analysis among populations, without discriminating the sex of the individuals. Different letters indicate significant differences between populations for the Tukey test at $p<0.05$. *Indicates significant differences between sexes for analysis of variance at $p<0.05$

Pedro Norte developed the lowest number of flowers, which suggests that flowering would be strongly dependent on available resources and environmental conditions.

Breeding systems

The results of the fertilization tests are summarized in Table 2 and confirm that the species is dioecious. Male flowers never formed fruits; although their gynoecium presented normal characteristics, it was sterile. Male flowers did not produce fruits either through self-fertilization, with pollen from their own anthers, or through cross-fertilization, with pollen from other flowers.

Female flowers produced fruit in a higher proportion when they were hand-pollinated (13/16–81.3%), and fruit production decreased when the flowers were exposed to the visit of natural pollinators (12/22–54.5%). The female flowers never self-fertilized.

Table 2 Pollination treatments in cultivated plants of *Gymnocalycium bruchii*

	Hand-pollination	Self-fertilization	Free-pollination
Male plants			
Formed flowers	8	18	37
Developed fruits	0	0	0
Female plants			
Formed flowers	16	12	22
Developed fruits	13	0	12

15 male and 15 female plants were studied, using 5 individuals per treatment. The flowers produced and the fruits developed were counted for each compatibility test

Floral anatomy and development

Detailed analysis of the ontogeny of the floral types revealed that both types of flowers begin development in the same way: the gynoecium completes its formation in both, while the androecium of the female flowers undergoes important modifications. The ontogeny of both floral types, separated into three stages (bud, pre anthesis and anthesis flowers) is detailed for a better understanding.

In the flower buds, the receptacle or pericarpel differs from the ovary wall in shape, arrangement, and compaction of its parenchymal cells; the receptacle also has secretory ducts that are at an initial stage of formation (Fig. 3A). The primordial ovules are inconspicuous (Fig. 3B), without a defined shape, with a megaspore mother cell, and a long funicle attached to the ovary wall by the placentas. The style is circular, has a stylar canal and numerous small vascular bundles, the parenchymal tissue that contains them and the tissue surrounding the canal are not yet differentiated (Fig. 3C). Towards the apex, style divides and forms numerous lobes (Fig. 3E), each lobule contains a small bundle and some secretory ducts in its early stages of development, on the surface, numerous papillae begin to differentiate (Fig. 3F). The androecium is formed by numerous anthers that at this stage are still very young, since they are closed (Fig. 3C), present a development tapetum on their wall and contain microspore tetrads (Fig. 3D).

At this inconspicuous stage of development, all flowers present identical characteristics in their reproductive organs; at the next stage, the flowers go through a process of sexual differentiation.

At pre-anthesis of the female flowers, the receptacle is further differentiated from the ovary wall (Fig. 4A) and the ovules have a developing embryo sac (Fig. 4B). At this stage, the anthers collapse just before their opening (Fig. 4C). This particular phenomenon occurs due to a constriction of the filament in its most apical zone, just below the insertion of the filament with the anther (Figs. 1I, J, 4D). The androecium becomes non-functional due to its indehiscent anthers and reabsorption of the microspores (Fig. 4E).

At the time of anthesis, the gynoecium is fully developed. The tissues of the ovary differ markedly; the receptacle has large secretory ducts, while the wall of the ovary is

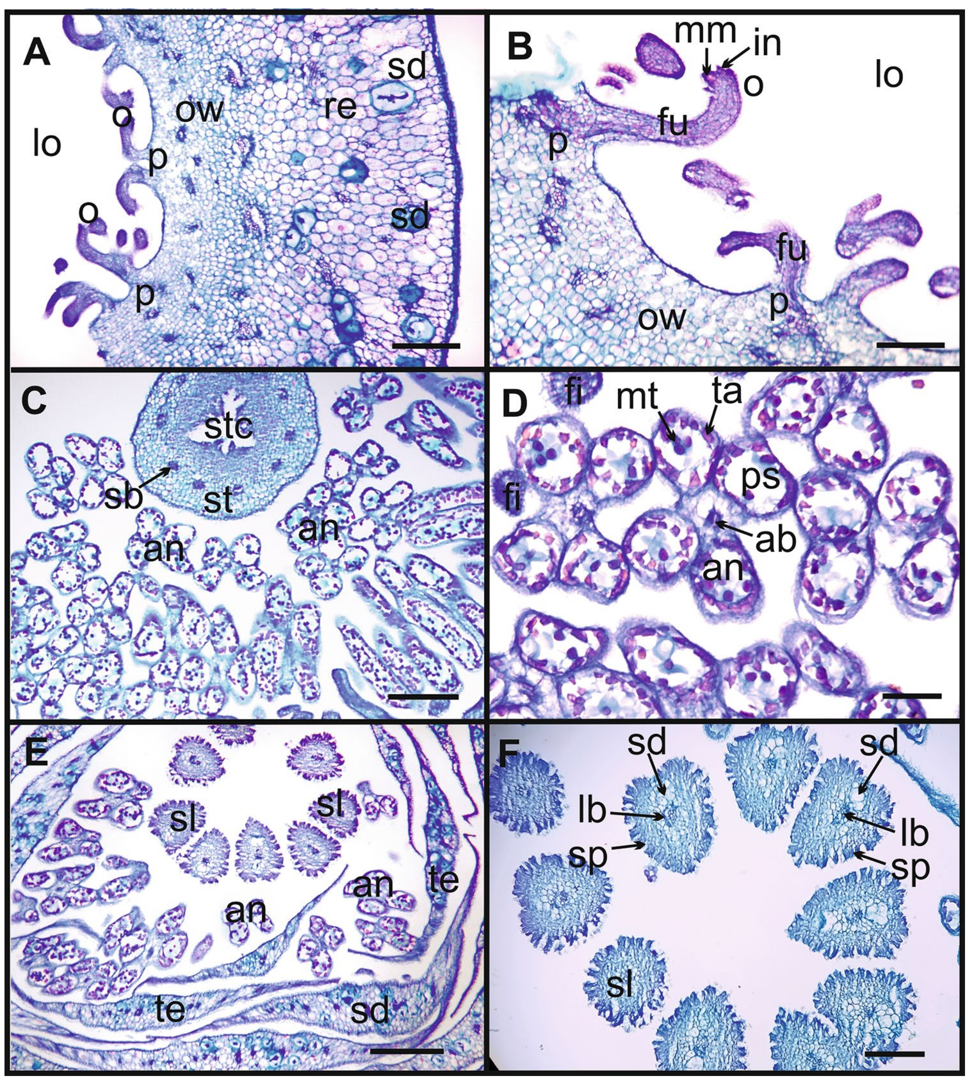

Fig. 3 Anatomy of the female flower bud. **A** ovary showing the ovary wall and the receptacle or pericarpel with some ducts. **B** numerous ovules with incipient development. **C** stile and indehiscents anthers. **D** young anthers with microspore tetrads and mature tapetum. **E** anthers and lobed stigma. **F** stigmatic lobules with small vascular bundles and incipient ducts. Abbreviations: ab: anther bundle, an: anther, fi: filament, fu: funiculus, in: integument, lb: lobel bundle, lo: loculus, mm: megaspore mother cell, mt: microspore tetrad, o: ovule, ow: ovary wall, p: placenta, ps: pollen sac, re: receptacle, sb: style bundle, sd: secretory duct, sl: stigma lobe, st: style, stc: style canal, te: tepal, ta: tapetum. Scales: **A**, **C**, **E** 30 µm. **B** 20 µm. **D**, **F** 10 µm

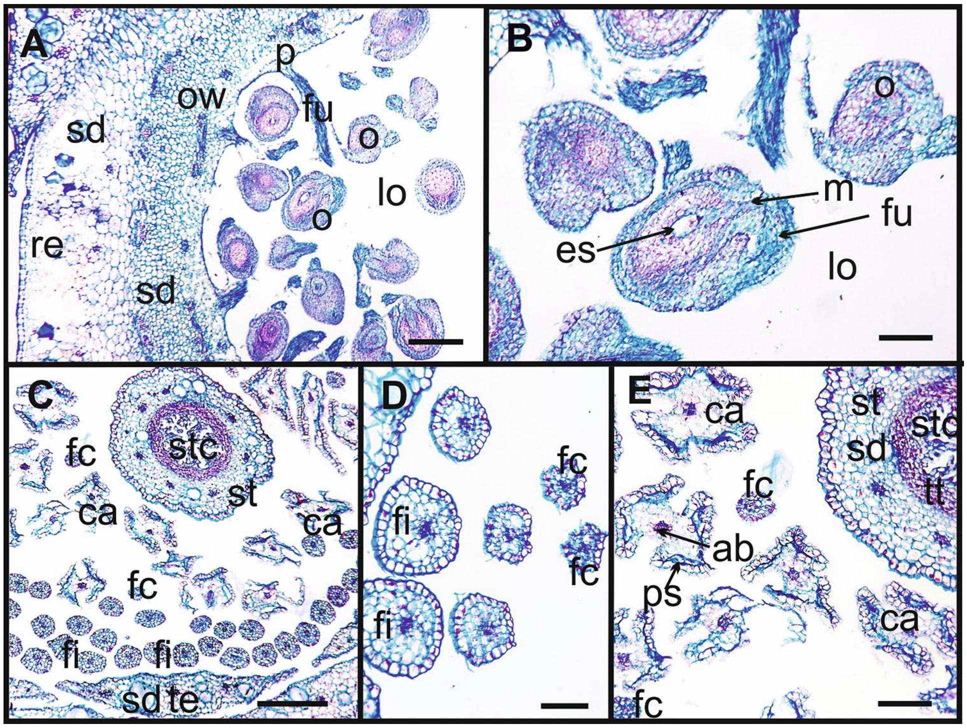

Fig. 4 Anatomy of female flower at pre-anthesis. **A** ovary with secretory ducts and numerous ovules. **B** ovule with immature embryo sac. **C** style and anthers collapse before anthesis. **D** detail of filaments at different heights, some with normal diameter and others at the height of contraction. **E** collapsed anthers and a filament with constriction. Abbreviations: ab: anther bundle, ca: collapsed anther, es: embryo sac, fc: filament constriction, fi: filament, fu: funiculus, in: integument, lo: loculus, m: micropyle, o: ovule, ow: ovary wall, ps: pollen sac, re: receptacle, sc: scale, sd: secretory duct, st: style, stc: style canal, te: tepal, tt: transmission tissue. Scale: **A**, **C**: 30 µm. **B**, **D**, **E**: 10 µm

composed of small and compact parenchyma cells (Fig. 5A). Both tissues, the receptacle and the wall of the ovary, are limited by a ring of collateral vascular bundles. Ovules are numerous, united by their long funicles, forming clusters, and at this stage, they present a conspicuous embryo sac (Fig. 5B). The ovules are campylotropous, bitegmic, crassinucellated, with a long funiculus. The nectary is located at the base of the floral tube, below the insertion region of the internal filaments to the perigonium. It presents a nectariferous tissue with several strata of small cells, with an evident nucleus and vacuole, innervated by abundant vascular tissue (Fig. 5C). The style has, in its basal portion, numerous bundles surrounded by undifferentiated parenchymal tissue (Fig. 5C); while, in the middle zone of the style, the bundles are completely surrounded by numerous secretory ducts (Fig. 5D). The style presents a large stylar canal covered by transmission tissue and an internal papillose epidermis (Fig. 5D). At its apex, the style divides into 7 or 9 stigmatic lobes (Fig. 5E), each containing a bundle and some related ducts (Fig. 5F); externally, numerous and long papillae or trichomes form the pollen receptive zone. At this stage, anthers are completely collapsed (Fig. 5D, E), and no microspores is found either within the pollen sacs or outside in the vicinity. The androecium is completely non-functional.

The male flower at pre-anthesis has numerous secretory ducts in the receptacle or pericarpel, with abundant mucilaginous content (Fig. 6A). The ovules present an immature embryo sac (Fig. 6B). The anthers are mature but still indehiscent (Fig. 6C), with pollen grains inside (Fig. 6C, D). The anther wall shows a developed endothecium and a consumed tapetum (Fig. 6D). The stigmatic lobules are almost fully developed, with abundant developing stigmatic papillae, a bundle and large secretory ducts (Fig. 6E).

At the time of anthesis, the ovary has the same characteristics as those of female flowers, with large secretory ducts in the outermost area of the receptacle (Fig. 7A). The ovules are mature, with the embryo sac developed and without signs of atrophy (Fig. 7B). The nectary showing a typical secretory nectariferous tissue, innervated by numerous vascular bundles and surrounded by a secretory epidermis, so that at this stage, when the male flowers are open, the nectary would be functional and would dump the nectar at the base of the floral tube (Fig. 7C). The style in its basal portion presents numerous small bundles and an evident stylar canal, but no secretory ducts or transmitting tissue are observed (Fig. 7C). In the middle zone of the style there is a large number of ducts accompanying the bundles and a very developed transmitting tissue surrounding the stylar canal (Fig. 7D). At the apex the style is divided into lobules, each containing a bundle and associated ducts, and is completely surrounded by abundant and long papillae (Fig. 7E). Morphoanatomically, the gynoecium shows no signs of malformations or infertility, so it is similar in all respects to the gynoecium of female flowers; however, it is non-functional, since they never produce fruit or seeds. The anthers are numerous (Fig. 7D) and develop normally until the moment of floral anthesis, when they open to release the pollen grains (Fig. 7F). Functionally, male flowers produce

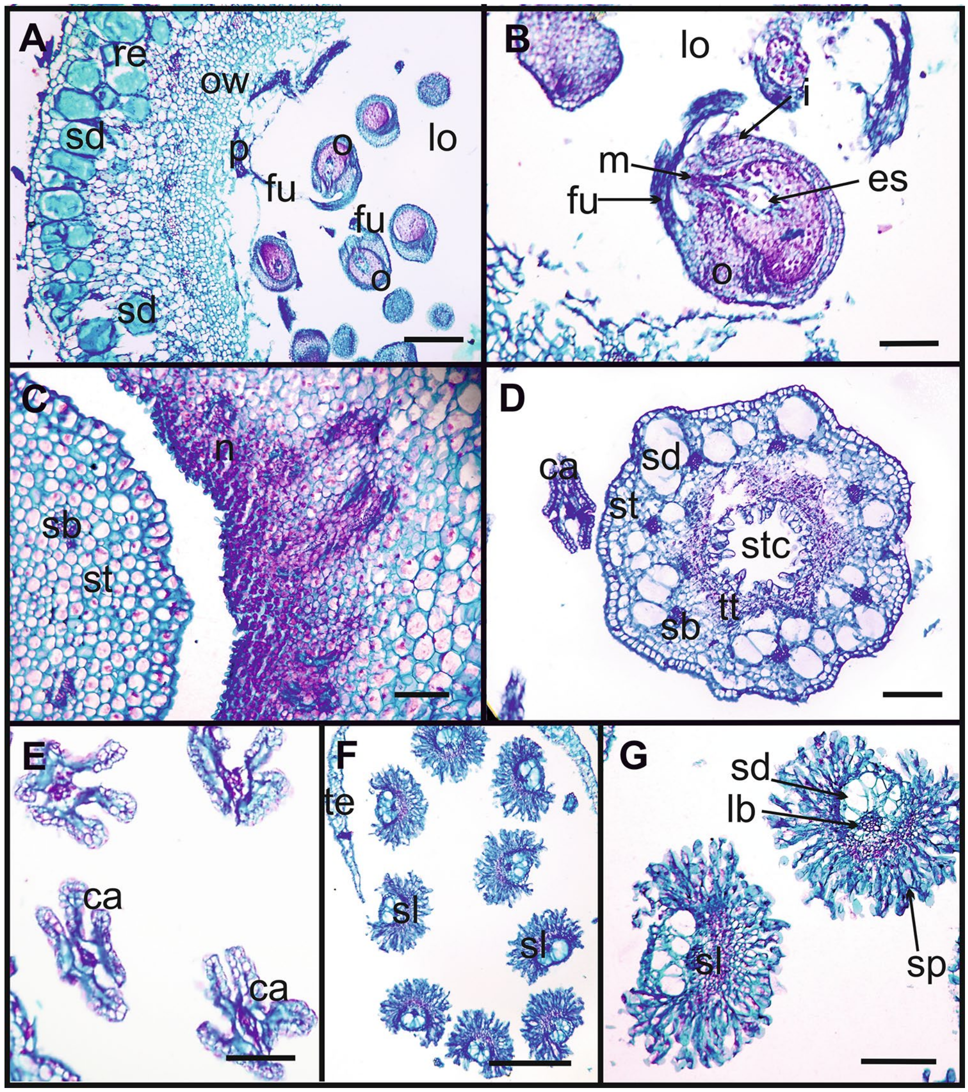

Fig. 5 Anatomy of female flower at anthesis. **A** Ovary showing receptacle with many secretory ducts full of mucilaginous content. **B** Detail of mature campylotropous ovule with embryo sac. **C** Nectariferous tissue located below the insertion region of the internal filaments. **D** One collapsed indehiscent anther and style with transmission tissue and style canal. **E** Collapsed anthers. **F** Stigma lobules and tepals. **G** Detail of papillose lobules, each one with one lobed bundle and secretory ducts. Abbreviations: ca: collapsed anther, es: embryo sac, fu: funiculus, i: integument, ld: lobe bundle, lo: loculus, m: micropyle, n: nectary, o: ovule, ow: ovary wall, p: placenta, re: receptacle, sb: style bundle, sd: secretory duct, sl: stigma lobe, st: style, stc: style canal, sp: stigmatic papillae, te: tepal, tt: transmission tissue. Scales: A: 30 µm. B, C: 10 µm. D, E, G: 20 µm. F: 50 µm

a large number of pollen grains, which can be found even inside the floral tube (Fig. 7C).

Discussion

Reproductive traits

To the best of our knowledge, this is the first study to analyze the breeding system in *Gymnocalycium,* and to comprehensively explore reproduction in a non-hermaphrodite cactus species. Our research confirms that *G. bruchii* is a dioecious species and presents two floral forms. The female flowers are smaller, bearing a functional gynoecium but abortive anthers. The male flowers are larger and have normal anthers but an unfuncional gynoecium and never form fruit. This type of reproductive system is rare in the family, since it has been reported in only ca. 15 species: four of *Pereskia*, three of *Consolea*, three of *Opuntia*, and five of *Echinocereus* (Sánchez and Vázquez-Santana 2018; Hernández-Cruz et al. 2018). The studied plants maintained the same sex for at least 6 years. This result is in agreement with findings of Díaz and Cocucci (2003), who reported a similar situation in *Opuntia quimilo* K. Schum, and indicates that sex may be genetically determined and not influenced by environmental conditions (Pannell 2017).

In *G. bruchii*, female flowers are easy to recognize, as in other dioecious, gynodioecious, or trioecious populations from other cactus species. These flowers have been identified based on the exomorphology of the non-functional androecium, which has small, indehiscent, dehydrated anthers, with malformed pollen or no pollen at all, and a constriction at the filament-anther union (Gutiérrez-Flores et al. 2017; Sánchez and Vázquez-Santana 2018; Hernández-Cruz et al. 2018; Callejas-Chavero et al 2021). In contrast, the male flowers were difficult to identify, since exomorphologically

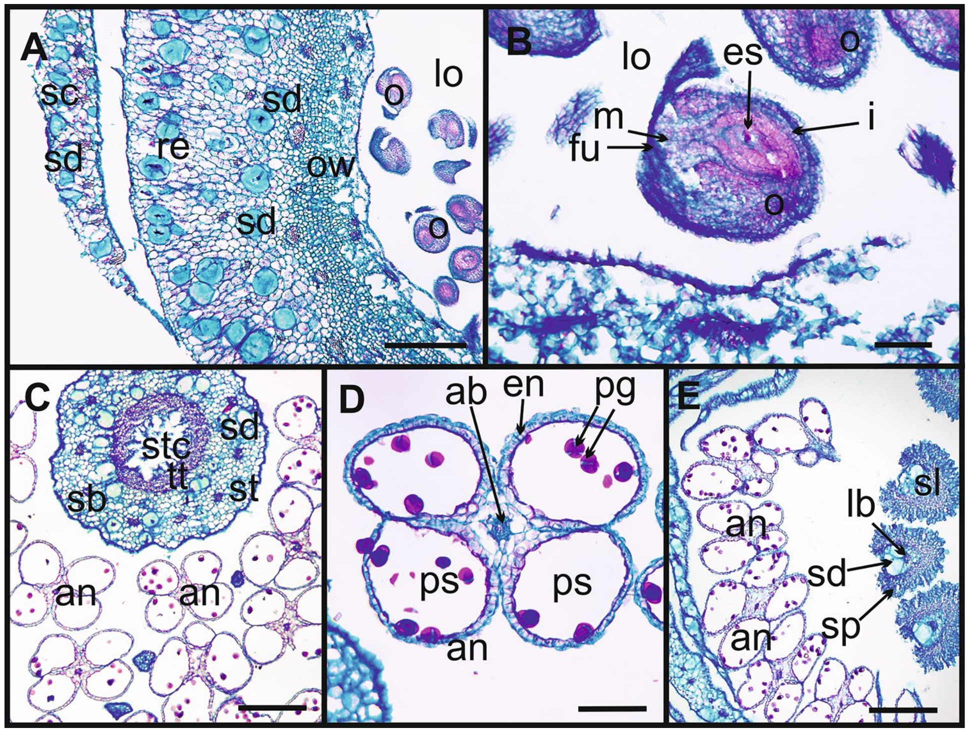

Fig. 6 Anatomy of male flower at pre-anthesis. **A** Ovary with numerous secretory ducts. **B** Ovule with immature embryo sac. **C** Style and numerous mature but indehiscent anthers. **D** Anthers with pollen grains, developed endothecium and consumed tapetum. **E** Stigmatic lobes and anthers. Abbreviations: an: anther, ab: anther bundle, en: endothecium, es: embryo sac, fu: funiculus, i: integument, lb: lobel bundle, lo: loculus, m: micropyle, o: ovule, ow: ovary wall, ps: pollen sac, pg: pollen grain, re: receptacle, sb: style bundle, sc: scale, sd: secretory duct, sl: stigma lobe, st: style, stc: style canal, sp: stigmatic papillae, tt: transmission tissue. Scales: **A**: 50 µm. **B**, D: 10 µm. **C**, **E**: 30 µm

they appeared to be hermaphrodite, i.e. both the gynoecium and androecium were normally developed, the dehiscent anthers were full of pollen grains and the gynoecium was fully mature, with its stigmatic lobes apparently receptive; however, such flowers never produced fruits and seeds. This feature is unexpected and has been described only in four dioecious species of *Echinocereus*; staminate flowers show a fully developed gynoecium with ovules capable of fertilization, but seeds are aborted at early stages of development (Hernández-Cruz et al. 2018).

Male flowers were consistently bigger than female ones, in both natural populations studied and also in the cultivated plants. The difference in flower size observed agrees with previous records in other cacti. In dioecious species, female flowers are smaller than male flowers (Hoffman 1992; Baker 2006), whereas in gynodioecious cacti female flowers are smaller than hermaphrodite ones (Fleming et al. 1994; Díaz and Coccuchi 2003; Sánchez and Vázquez-Santana 2018; Callejas-Chavero et al. 2021). We hypothesize that these patterns could be a result of strong competition: male flowers, responsible for the production of viable pollen, may invest more resources to attract pollinators, whereas female flowers have an extra cost in producing seeds, so they can compensate for this cost by generating smaller flowers, as observed by Díaz and Cocucci (2003).

In the field, we found a higher proportion of plants with male flowers, as well as plants with no flowers at all. While it is known that flowering and fruiting may differ between years due to internal or external factors (Koenig 2021), in these slow-growing cacti, female plants may not flower every year, since fruit and seed formation involves a significant energetic cost to the plant (Wilson and Harder 2003; Wheelwright and Logan 2004). In *Wigginsia sessiliflora* (Hook) D. M. Porter, another globose cactus species with a low growth rate, 40% of reproductively mature plants did not produce fruit in a given year (Ceballos et al. 2022). In addition, our results show that environmental conditions strongly influence the development and reproduction of *G. bruchii.* Thus, in San Pedro Norte, individuals were smaller and produced fewer flowers, both male and female, which translated into low reproduction.

Breeding systems

Fertilization tests confirmed that *G. bruchii* is dioecious and strongly dependent on the agency of pollinators. Male flowers never formed fruit, regardless of the pollen source. The female flowers never formed pollen, so they were not able to self-fertilize, but they formed fruits and seeds if they were pollinated with pollen from male flowers. These types of compatibility tests have only been performed on four dioecious species of *Echinocereus*, and it was also confirmed that only the functionally female flowers form viable fruits and seeds (Hernández-Cruz et al. 2018). Dioecious species (Hernández-Cruz et al. 2018), as well as taxa that have hermaphroditic but self-incompatible flowers (Delgado-Ramírez et al. 2021), need the effective visit of pollinators for reproduction.

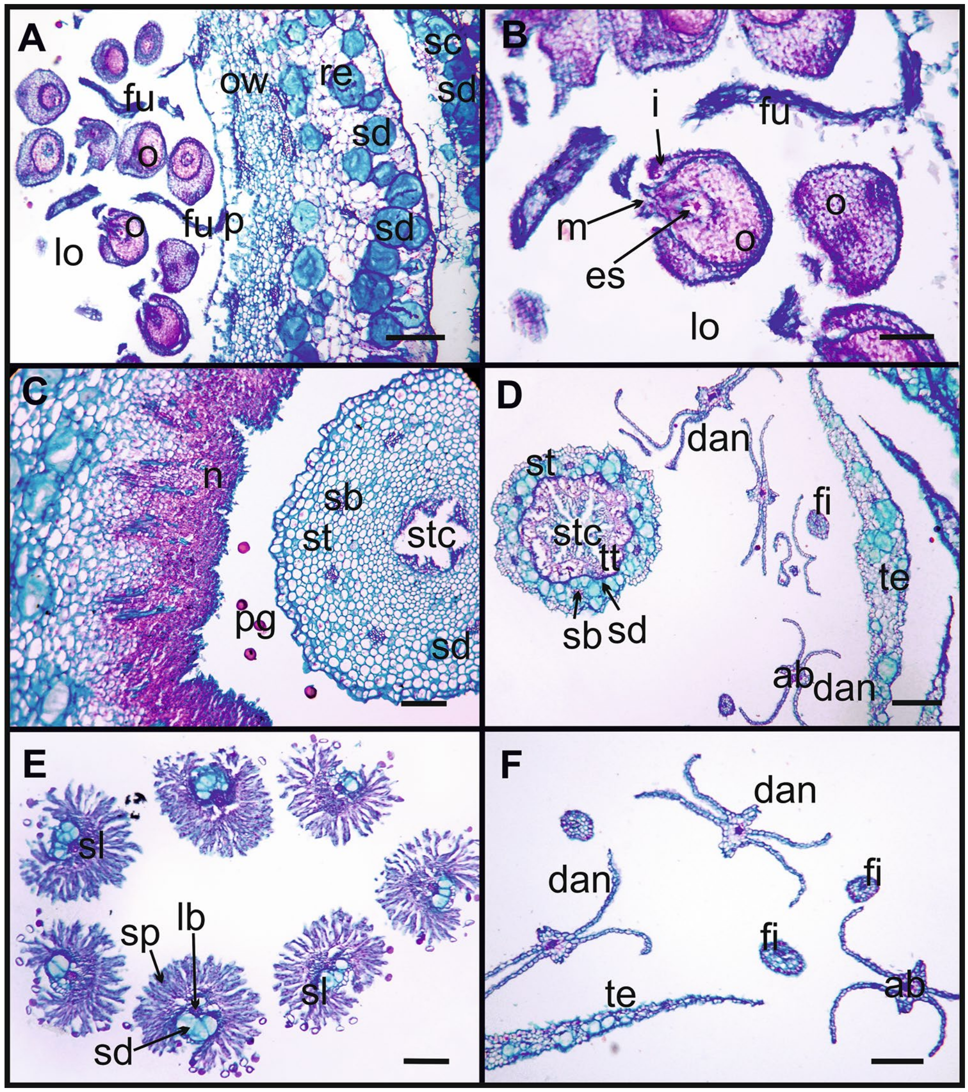

Fig. 7 Anatomy of male flower at anthesis. **A** Ovary with numerous ovules and receptacle with many ducts full of mucilaginous content. **B** Detail of mature campylotropous ovule with embryo sac. **C** Nectary located below the insertion region of the internal filaments. **D** Tepals, mature and dehiscent anthers, and style with transmission tissue and style canal. **E** Detail of papillate stigmatic lobes. **F** Detail of dehiscent anthers. Abbreviations: ab: anther bundle, dan: dehiscent anthers, es: embryo sac, fi: filament, fu: funiculus, i: integument, lo: loculus, m: micropyle n: nectary, o: ovule, ow: ovary wall, p: placenta, pg: pollen grain, re: receptacle, sb: style bundle, sd: secretory duct, sc: scale, sl: stigma lobe, st: style, sp: stigmatic papillae, stc: style canal, te: tepal, tt: transmission tissue. Scale: A, D, F: 30 µm. B: 10 µm. C, F: 20 µm

By contrast, in gynodioecious species all flowers set fruits and seeds, indicating that the female function remains active in both flower types. However, in *M. magnimamma* Haw (Callejas-Chavero et al. 2021), hermaphroditic flowers form fewer, smaller seeds, and have a lower germination rate than female ones, whereas in *Opuntia robusta* H. L. Wendl. ex Pfeiff., hermaphroditic flowers present lower viability (Del Castillo and Trujillo-Argueta 2009). None of these characteristics were observed in *G. bruchii* so we categorically rule out this intermediate compatibility system.

Floral development

Although studies of the anatomy and development of flowers and fruits are common in Cactaceae (Fernández et al. 2020), few studies include non-hermaphrodite species (Strittmatter et al. 2002, 2006, 2008; Flores-Rentería et al. 2013; Sánchez and Vázquez-Santana 2018; Hernández-Cruz et al. 2018, 2019; Ramadoss et al. 2022). Ontogenetic studies of gynodioecious or dioecious species are essential to know the precise timing of sexual differentiation and the underlying mechanisms.

In *G. bruchii*, female flowers show normal anatomy and development of the gynoecium and forms viable fruits and seeds after fertilization. On the other hand, the anthers collapse at the time of formation of the microspore tetrads. This observation coincides with records in dioecious species of *Echinocereus* (Hernández-Cruz et al. 2018), *Opuntia* (Flores-Rentería et al. 2013; Hernández-Cruz et al. 2019), and *Consolea* (Strittmatter et al. 2002, 2006, 2008), in which the microspore mother cells cease to develop as a result of degeneration of the tapetum. Premature expression of programmed cell death (PCD) in tapetum cells during microsporogenesis is the major factor leading to male sterility in species of *Echinocereus* (Hernández-Cruz et al. 2018).

The male flowers of *G. bruchii* present normal anthers that produce viable pollen grains. On the other hand, the gynoecium presents identical morphological and anatomical characteristics to those of the gynoecium and ovules of female flowers; however, it never forms fruits and seeds, so it is functionally sterile. This particular phenomenon is rare in Cactaceae, since in male flowers normal development of the gynoecium does not usually occur and this organ presents morphoanatomical alterations (Orozco Arroyo et al. 2012; Flores-Rentería et al. 2013). This has only been reported with the same characteristics in dioecious species of *Echinocereus* and *Consolea* genera, in which the gynoecium is normal with stigma, style, and numerous apparently normal ovules, but does not form viable seeds and fruits (Strittmatter et al. 2008; Hernández-Cruz et al. 2018). In the staminate flowers of Consolea, ovules undergo early senescence due to the degeneration in the nucellus and the embryo sac. Thus ovule receptivity in staminate versus carpellate floral forms has been attributed to heterochrony in the ovule developmental programme (Strittmatter et al. 2008). In *Opuntia robusta* (Hernández-Cruz et al. 2019) the ovules either do not complete their development or develop but then abort, and evidence of Programmed Cell Death (PCD) was detected in both the placenta and the ovules. In species of *Echinocereus* (Hernández-Cruz et al. 2018), the ovules are fertilized, but the seeds abort at the early stages of embryogenesis. Evidence of PCD was observed in the seed-nourishing tissues, funiculus, and nucellus; hindering proper seed development.

In *G. bruchii*, the mechanisms that determine how and why fruits and seeds fail to develop in male flowers and anthers collapse before opening in female flowers, resulting in functionally unisexual flowers, are still unclear and require specific studies. It is likely that the genetic and molecular mechanisms involved in their sexual differentiation would be similar to those reported for other dioecious Angiosperms species (Wu and Cheung 2000; Leite Montalvão et al. 2021) and some non-hermaphrodite species of Cactaceae, in which temporal and spatial changes in the patterns of PCD were proposed to be mainly responsible (Strittmater et al. 2008, Flores-Renteria et al. 2013, Hernández-Cruz et al. 2018, Hernández-Cruz et al. 2019, Ramadoss et al. 2022). In non-hermaphrodite species, male infertility is caused by the disintegration of the tapetum before microspore maturation. The disintegration is caused by the presence of mutations in some of the genes controlling anther development, leading to premature and displaced PCD, as evidenced by the detection of DNA fragmentation of the component cells. Tapetal disintegration to achieve male sterility could be a conserved trait of Cactaceae (Flores-Renteria et al. 2013; Hernández-Cruz et al. 2018; Ramadoss et al. 2022). On the other hand, ovule or seed abortion is the main cause for female sterility in most species, and in general it was observed that female sterility would be related to a premature PCD activation, since DNA fragments and abnormal cell collapse and disintegration were detected in several areas in and around the ovule. Female sterility appears to be a more complex process, and due to scarce studies, the role of PCD is not clearly determined and it is not know if this mechanism is conserved throughout the family (Hernández-Cruz et al. 2018, 2019; Ramadoss et al. 2022).

In *G. bruchii* the use of molecular and genetic techniques, such as TUNEL assays method for detecting DNA fragmentation, which is a hallmark of PCD, would be an ideal tool for the understanding of its sexual differentiation. In addition, the implementation of such techniques in comparative analyses between hermafrodite and non-hermaphrodite species could help to better assess the underlying genetic basis redeployed in the evolution of reproductive mechanisms in the genus.

Phylogenetic, ecological, and conservation considerations

In angiosperms, the dominant sexual system is hermaphroditism, while dioecy is unusual (only 6%) (Muyle et al. 2020). The evolution of dioecy from hermaphroditic ancestors required at least two evolutionary steps; therefore, systems or combinations such as gynodioecy, androdioecy, subdioecy, or trioecy can be considered intermediate stages (Charlesworth 2006; Muyle et al. 2020). The genus *Gymnocalycium* can serve as a clear model, since it is well-inserted within the Cactoideae and presents important evolutionary radiation (Meregalli et al. 2010; Demaio et al. 2011). While most species have been described as hermaphroditic, putative gynodioecious populations were observed (Gurvich, unpublished data) and herein we confirm that *G. bruchii* is dioecious.

Breeding systems influence various aspects of a population. Dioecious species have high genetic variability due to cross-fertilization (Charlesworth 2006); thus, genetic variability is associated with more efficient purifying selection and greater adaptive evolution (Muyle et al. 2020). On the other hand, population characteristics are strongly influenced by ecological factors such as size, sex ratio, and the presence of pollinators to ensure cross-fertilization and seed dispersal (Charlesworth 2006). In dioecious species, reproduction only occurs if pollen is passed from males to females; therefore, pollen limitation is one of the main factors restricting seed formation. This trade-off would be compensated by a combination of several floral traits, which attract small animals such as insects in search of rewards (Ohya et al. 2017), and with an allocation of resources to reproduction at the expense of growth, especially of female plants (Charlesworth 2006). Finally, small environmental variations, such as nutrient or water availability, can lead to marked differences among individuals within a given population (Buhanan and Briggs 2010).

The interdisciplinary study of dioecious species, such as *G. bruchii,* is essential for a holistic understanding of the dynamics of the species and its interaction with the surrounding environment. However, understanding the evolution of floral traits of the entire genus is also essential. Furthermore, taking into account that at least 30% of cactus species are at risk of extinction due to human activities and global change (Goettsch et al. 2015; Gurvich 2019), more information on basic aspects of species and populations, especially the study of reproductive biology and ecological morphoanatomy, is necessary to develop efficient conservation practices and obtain a better understanding of the impact of imminent climate change on natural populations.

Author contribution statement ND and DG designed the study. ND, DG, NA, SG and ML performed the field observations and experiments on cultivated plants. ND and NA carried out the laboratory work. ND, DG and RS wrote the manuscript. All authors have read and agreed to the published version of the manuscript.

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Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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